

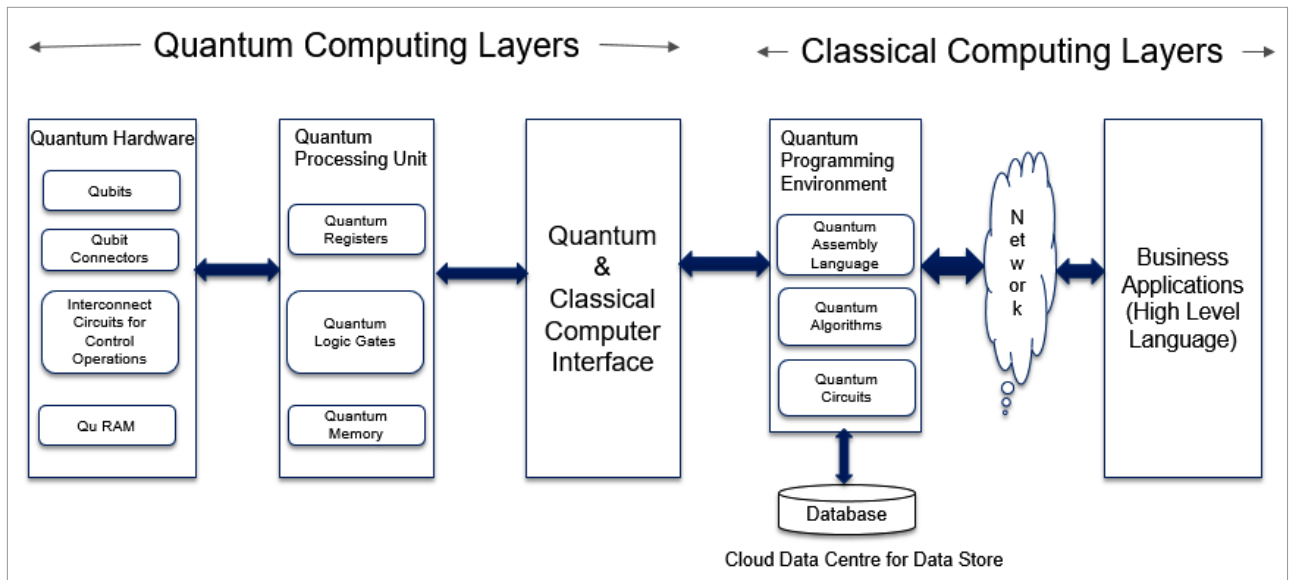


Figure 1: QCP architecture

Programmers access production-ready quantum software remotely for development and testing.

Quantum algorithms: Programmers leverage quantum algorithms to define the computing task. Algorithms help in gaining speed and communicating with other computing tasks that are running on the QCP. Programmers must identify or design suitable quantum algorithms that can solve the problem on hand.

Non-portability of software: Quantum computing platforms are currently evolving and the software developed by platform owners is native in nature. This software follows its own standards, with proprietary programming APIs and tools. Examples are Qiskit from IBM, Quantum Development Kit from Microsoft, and Cirq from Google.

QCP architecture

A quantum computing platform consists of two layers – the quantum computing layer and the classical computing layer (Figure 1).

The quantum computing layer is composed of three main elements: quantum hardware, the quantum processing unit (QPU), and the quantum-classical interface. Quantum hardware includes qubits, typically

realised using superconducting loops, along with interconnect circuitry that enables its control and operation. The QPU encompasses quantum registers, quantum logic gates, and quantum memory, forming the core of quantum computation. The quantum-classical interface, comprising both hardware and software, facilitates communication and interaction between classical computers and the QPU. Complementing this is the classical computing layer, which consists of the quantum programming environment, cloud data centres, and business applications that together support the practical deployment and utilisation of quantum computing.

The quantum programming environment consists of several key components that enable interaction with quantum systems and the execution of algorithms. It includes quantum assembly language for instructing the QPU, high-level programming languages for writing quantum programs, and quantum algorithms designed to solve complex problems more efficiently than classical computers. Quantum circuits serve as the standard model for representing quantum computation, where each step of an algorithm is expressed as a sequence of quantum logic gates that transform input qubits through matrix and vector

operations. A high-level programming API provides the necessary instructions for composing quantum programs, which typically involve tasks such as mapping classical bits to qubits, initialising qubit states, constructing quantum circuits with suitable logic gates to represent algorithmic steps, and performing measurements to obtain output qubit states that are then translated back into classical bits. Finally, a cloud data centre is used to store and manage processing data derived from the outputs of quantum algorithms.

Quantum computing technology landscape and its applications

Various industry-leading organisations are working on quantum computing. IBM, Microsoft, Alibaba, D-Wave Systems, Google, Nokia, Fujitsu and Intel continue to work and compete for authority in this field. The attempt is to deliver seamless data encryption, real conversations with AI, and better financial modelling through quantum technology.

IBM: IBM has been working to develop a quantum computer for over 35 years. In December 2023 it unveiled a 1121-qubit quantum processor called Condor, setting a new industry benchmark for qubit count.

IBM Quantum Experience is a cloud